

GROUP 3: HEALTHCARE

Sub-Team	Roster
A	Dherick Salvacion, David Bautista & Steve Jacinto
B	Rodesa Padua & Harvey Sierra
C	Grant Solema, Jim Daryll Campbell, & Jonel Benedict Orada
D	Carlo Gepiga & Justin Canon

Section 1: Human Resource Snippet: Finance Payload Snippet (json).

```
{ } hr_payroll_payload.json > ...
1  {
2    "employeeId": "EMP-2026-00123",
3    "payPeriod": "2026-08-16 to 2026-08-31",
4    "grossPay": 35000.00,
5    "taxDeduction": 4200.00,
6    "netPay": 30800.00
7  }
```

Finance Payload Snippets (json).

```
} finance_ledger_payload.json > ...
1  {
2    "journalId": "JE-2026-00987",
3    "debitAccount": "Payroll Expense",
4    "creditAccount": "Cash/Bank",
5    "totalDisbursement": 30800.00
6  }
```

GROUP 3: HEALTHCARE

Section 2: Cross-Module Transformation & Transaction Logic. The transformation matrix maps HR payroll fields to Finance/Accounting journal-entry fields.

INTEGRATION MAPPING TABLE

Source Field (HR Module)	Target Field (Finance Module)	Data Type	Transformation	Accounting / Business Logic Rule
grossPay	Debit_Amount	Currency / Float	Convert grossPay to currency value	Debits the Payroll Expense Account (Code 5001)
taxDeduction	Credit_Amount	Currency / Float	Convert taxDeduction to currency value.	Credits the Tax Liability Account (Code 2010).
netPay	Credit_Amount	Currency / Float	Convert netPay to currency value.	Credits the Cash Disbursement Account (Code 1010)

Transactional Resiliency Check (2PC / ACID-Style Rollback)

The integration uses a 2-Phase Commit (2PC) / ACID-style transaction. First, HR marks the payroll transaction as PENDING and sends the payload to Finance. Finance validates that the journal entry will balance and stages it without finalizing it. After Finance confirms that the entry is valid, the transaction enters Phase 2 (Commit): HR changes the status to POSTED while Finance finalizes the ledger entry. If the network drops before the commit is completed, a rollback is triggered. HR returns the payroll status to APPROVED_NOT_POSTED and Finance discards the staged entry. This ensures that either both systems update successfully or neither system records the transaction.

GROUP 3: HEALTHCARE

Section 3: Transactional & Security Architecture.

3.1 Authentication — OAuth 2.0 Bearer Token

The HR system uses an OAuth 2.0 Bearer Token to securely communicate with the Finance system. Before sending payroll information, the HR system authenticates and receives a valid access token. The Finance system verifies the token before accepting the payroll request. This helps prevent unauthorized systems from accessing the Finance API.

3.2 Audit & Compliance

Audit Log Example:

```
{
  "auditId": "AUD-2026-00456",
  "timestamp": "2026-09-08T09:00:06+08:00",
  "actorSystem": "HRPayrollModule",
  "action": "POST_LEDGER_ENTRY",
  "journalId": "JE-2026-00987",
  "authMethod": "OAuth2.0-BearerToken",
  "result": "SUCCESS"
}
```

Audit Log Explanation:

The audit log records important information about each HR-to-Finance transaction, including the audit ID, date and time, system responsible for the action, action performed, journal ID, authentication method, and result. This allows authorized personnel to trace and review financial transactions for auditing and compliance purposes.